

# **ECE321**

## **Electronic circuits**

### **Lecture (05)- Op. Amp\_nonlinear\_Apps**

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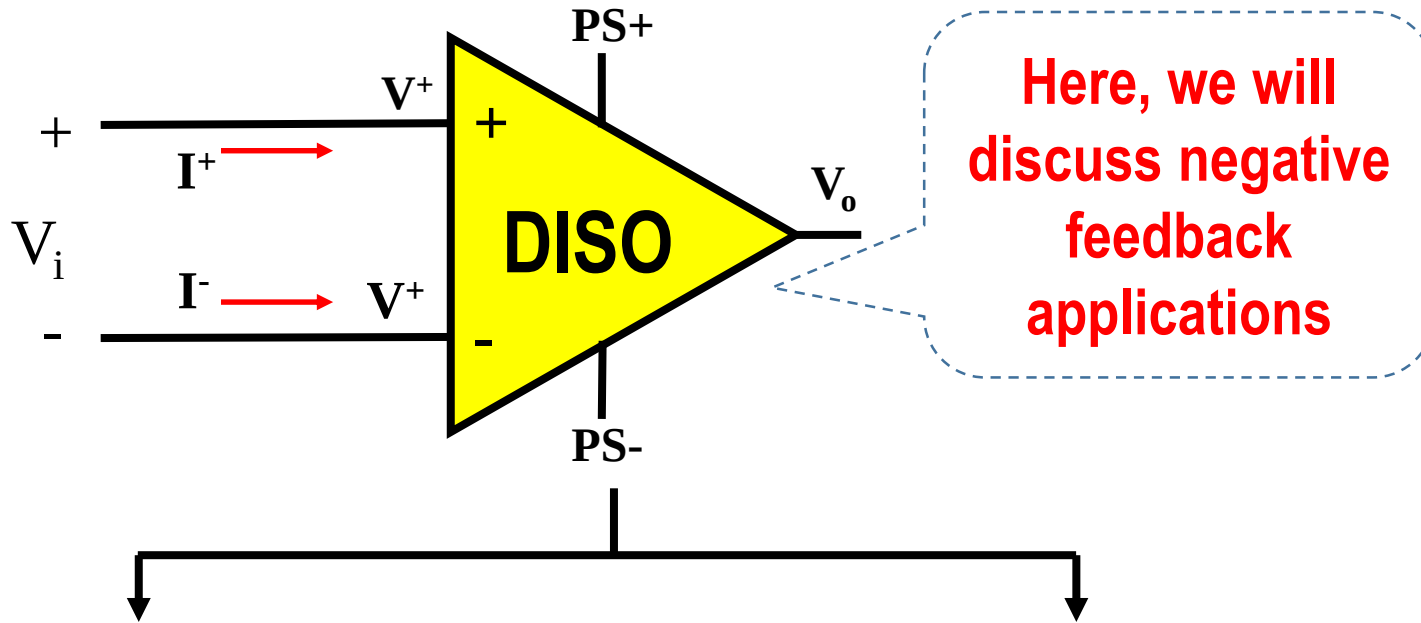


- ☐ Applications of Op amp
- ☐ Non-linear Applications
  - ☐ Logarithmic (log) Amplifier
  - ☐ Anti-Logarithmic (anti-log) Amplifier
- ☐ Exercise



- ☐ **Applications of Op amp**
- ☐ **Non-linear Applications**
  - ☐ **Logarithmic (log) Amplifier**
  - ☐ **Anti-Logarithmic (anti-log) Amplifier**
- ☐ **Exercise**

# Applications of Op amp

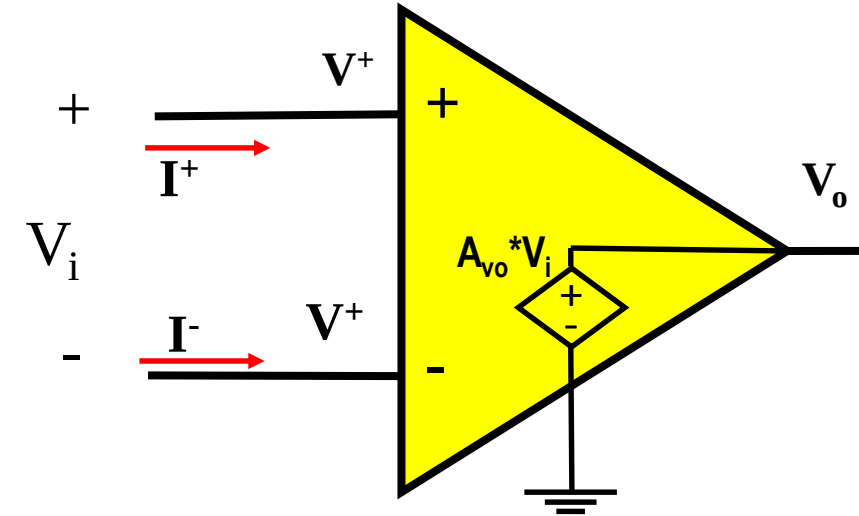


## Linear Applications

- ☐ Inverting amplifier
- ☐ Inverting summing amplifier
- ☐ Inverting integrator amplifier
- ☐ Inverting differentiator amplifier
- ☐ Non-inverting amplifier
- ☐ Voltage follower amplifier
- ☐ Difference amplifier

## Non-linear Applications

- ☐ Log Amplifier
- ☐ Anti-log Amplifier



## For idea DISO amplifier

$$I^+ = I^- = 0$$

$$V^+ = V^-$$

$$V_o = A_{VO}(V^+ - V^-)$$

$$Z_i = \infty$$

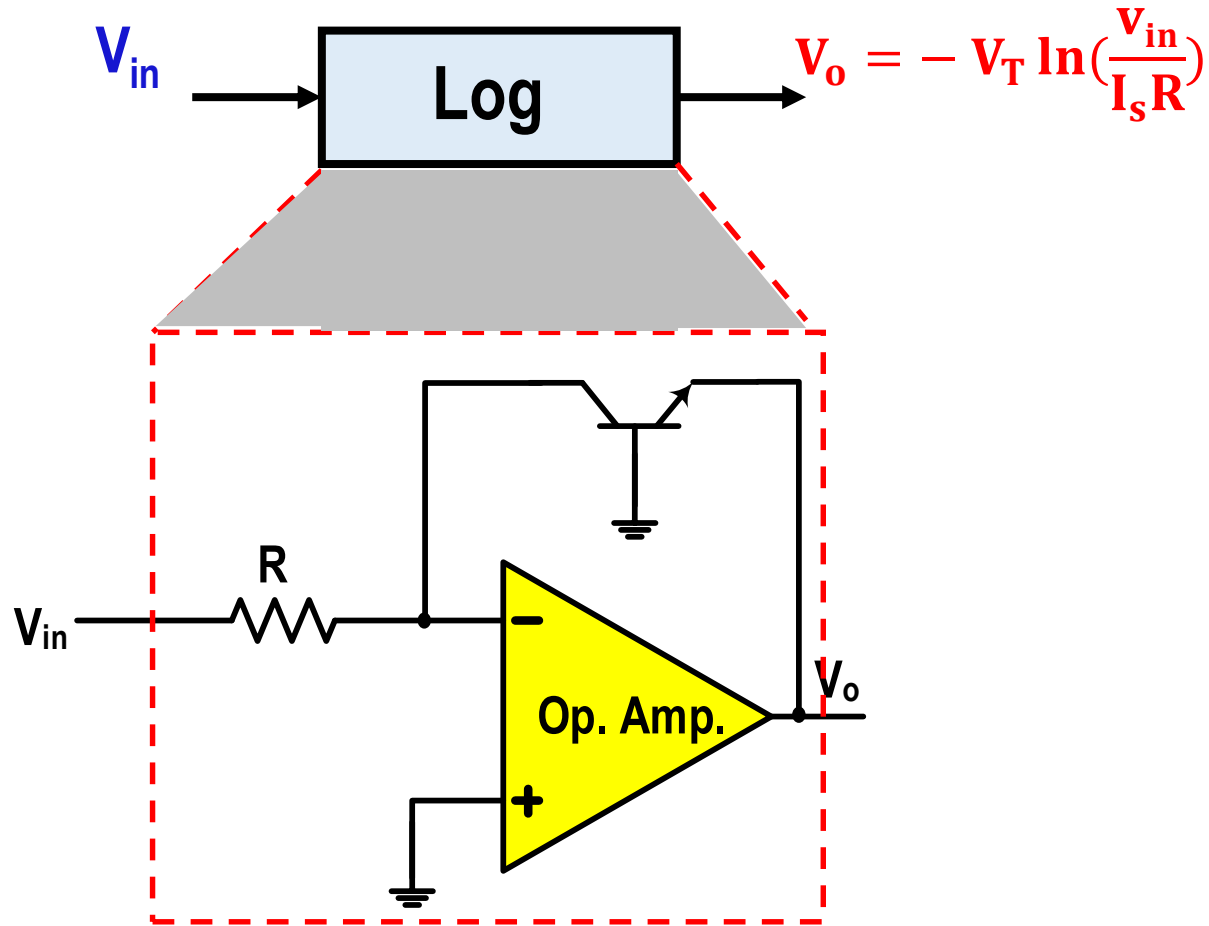
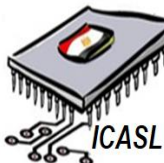
$$Z_o = 0$$

$$A_{VO} = \infty$$



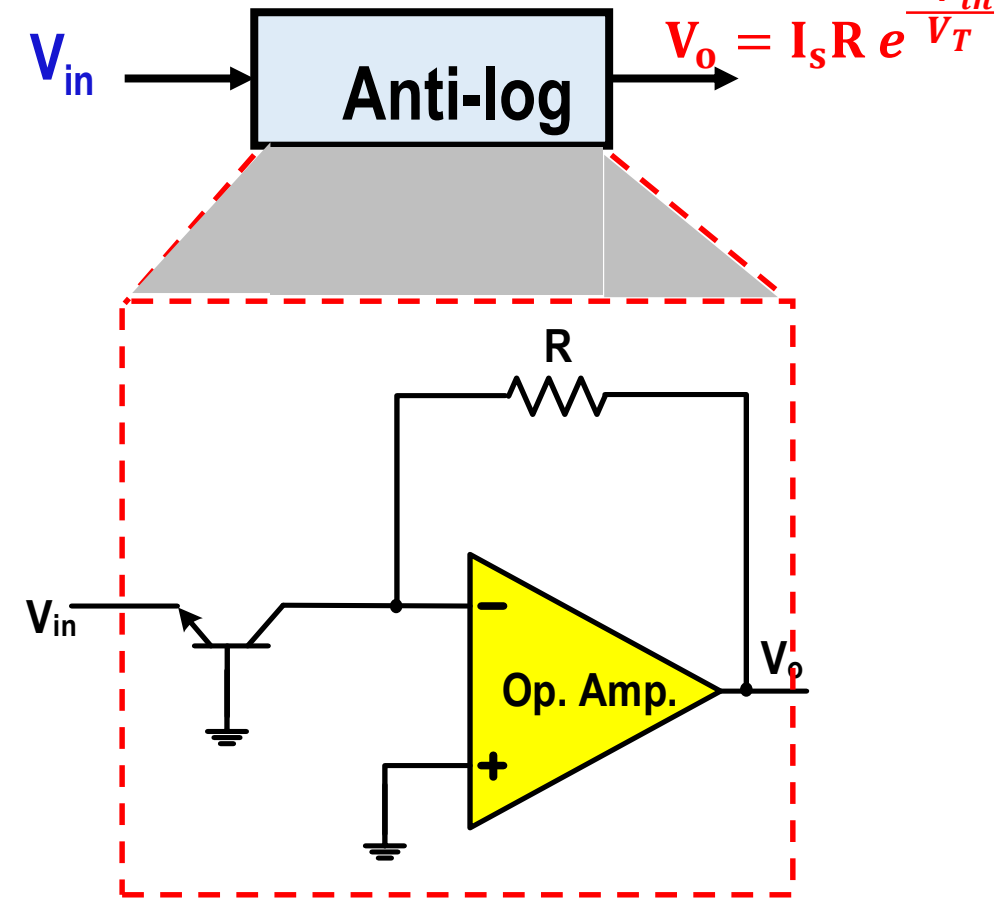
- ☐ Applications of Op amp
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# Non-linear Applications (1/2)



$V_{in}$  should be +Ve  
 $V_o$  should be -Ve  
 $I_s * R$  is dependent on temperature

Can be used to implement  
 non linear functions  
 $V_o = V_1 * V_2$ ,  $V_o = V_1 / V_2$ ,  $V_o = \sqrt{V_1}$   
 ,.....etc



$V_{in}$  should be -Ve  
 $V_o$  should be +Ve  
 $I_s * R$  is dependent on temperature

# Non-linear Applications (2/2)



□ Note:

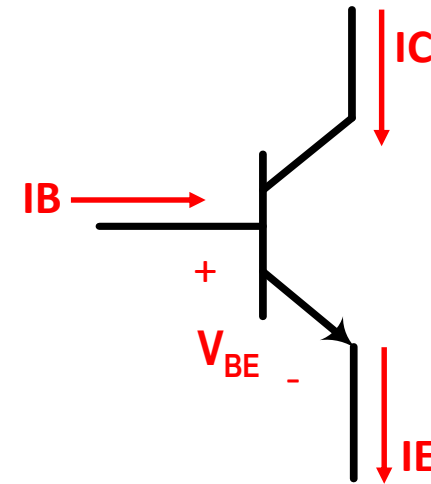
$$\ln(V1 * V2) = \ln(V1) + \ln(V2)$$

$$\ln\left(\frac{V_1}{V_2}\right) = \ln(V1) - \ln(V2)$$

$$\ln(V1 * V_2^2) = \ln(V1) + 2 * \ln(V2)$$

$$\log_e V_1 = \ln(V1)$$

$$e^{\ln(V_1)} = V_1$$



$$I_C = I_S \left( e^{\frac{V_{BE}}{V_T}} - 1 \right) \quad @V_{BE} \gg 3V_T$$

$$I_C \cong I_S \left( e^{\frac{V_{BE}}{V_T}} \right)$$

$$V_{BE} = V_T \ln\left(\frac{I_C}{I_S}\right)$$

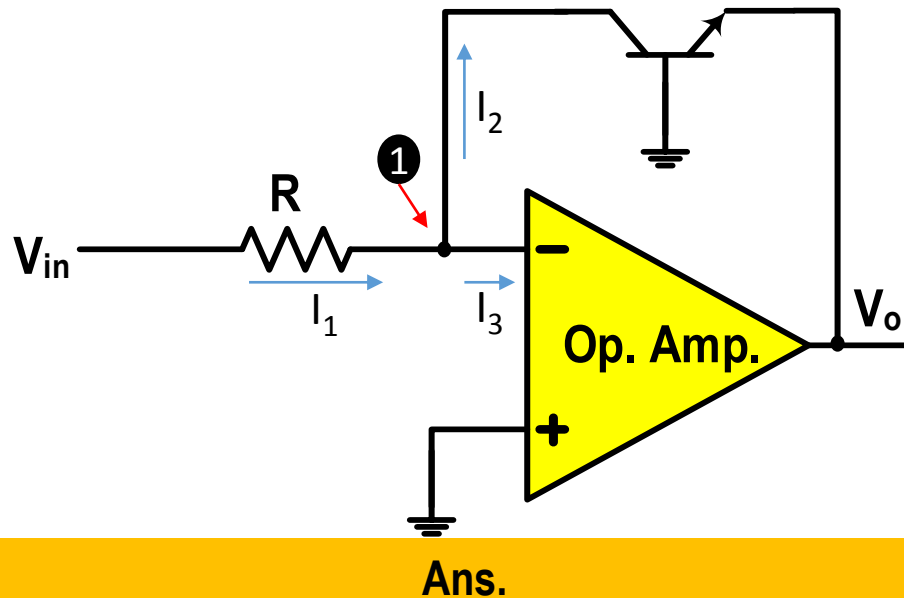


- ❑ Applications of Op amp
- ❑ Non-linear Applications
  - ❑ **Logarithmic (log) Amplifier**
  - ❑ Anti-Logarithmic (anti-log) Amplifier
- ❑ Exercise

# Logarithmic (log) Amplifier



□ Find Expression of  $V_o(V_{in})$



For ideal op amp

$$I^+ = I^- = 0$$

$$V^+ = V^- = 0$$

Take node (1)

$$I_1 = I_2 + I_3$$

$$\frac{V_{in} - 0}{R} = I_C$$

where  $I_C \cong I_S \left( e^{\frac{V_{BE}}{V_T}} \right)$

$$\frac{V_{in} - 0}{R} = I_S \left( e^{\frac{V_{BE}}{V_T}} \right)$$

where  $V_{BE} = V_B - V_E$   
 $= 0 - V_o$   
 $V_{BE} = -V_o$

$$\frac{V_{in}}{R} = I_S \left( e^{\frac{-V_o}{V_T}} \right)$$

$$\frac{V_{in}}{I_S R} = \left( e^{\frac{-V_o}{V_T}} \right)$$

$$\ln\left(\frac{V_{in}}{I_S R}\right) = \ln\left(e^{\frac{-V_o}{V_T}}\right)$$

$$\ln\left(\frac{V_{in}}{I_S R}\right) = \frac{-V_o}{V_T}$$

$$V_o = -V_T \ln\left(\frac{V_{in}}{I_S R}\right)$$

$V_{in}$  should be +Ve

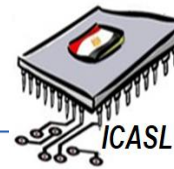
$V_o$  should be -Ve

$I_S * R$  is dependent on temperature

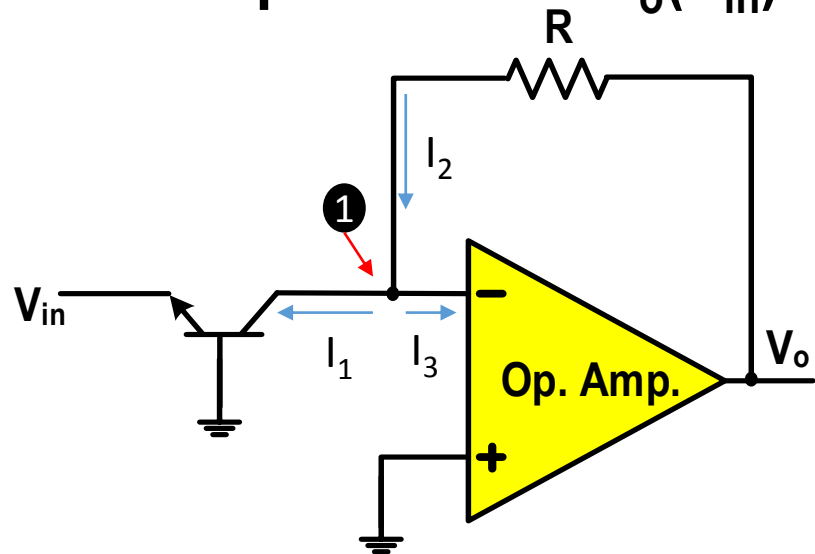


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# Non-linear Applications



□ Find Expression of  $V_o(V_{in})$



Ans.

For ideal op amp

$$I^+ = I^- = 0$$

$$V^+ = V^- = 0$$

Take node (1)

$$I_2 = I_1 + I_3$$

$$\frac{V_o - 0}{R} = I_c$$

$$\frac{V_o - 0}{R} = I_s \left( e^{\frac{V_{BE}}{V_T}} \right)$$

where  $I_c \cong I_s \left( e^{\frac{V_{BE}}{V_T}} \right)$

where  $V_{BE} = V_B - V_E$   
 $= 0 - V_{in}$   
 $V_{BE} = -V_{in}$

$$V_o = I_s R \left( e^{\frac{-V_{in}}{V_T}} \right)$$

$V_{in}$  should be -Ve

$V_o$  should be +Ve

$I_s * R$  is dependent on temperature



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# Exercise (1/5)



□ Ex.1, Design using Op. Amp a circuit to realize the following function

(a)  $V_o = K \sqrt{V_1}$ , where  $K$  is constant

Ans.

## 1. Refine the function

$$V_o = k\sqrt{V_1}$$

$$V_o \propto \sqrt{V_1}$$

$$V_o \propto V_1^{1/2}$$

$$\ln(V_o) \propto \frac{1}{2} \ln(V_1)$$



You have to do  
3X steps

$$= K * \sqrt{V_1}$$

$$= \sqrt{I_S R} * \sqrt{V_1}$$

$$= I_S R * \left(\frac{V_1}{I_S R}\right)^{1/2}$$

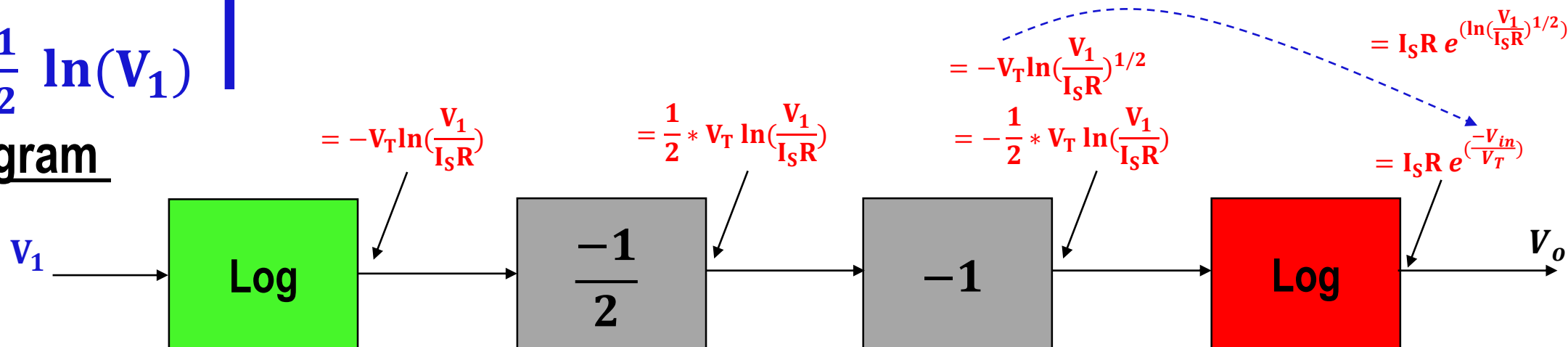
$$= I_S R e^{(\ln(\frac{V_1}{I_S R})^{1/2})}$$

$$= -V_T \ln\left(\frac{V_1}{I_S R}\right)^{1/2}$$

$$= -\frac{1}{2} * V_T \ln\left(\frac{V_1}{I_S R}\right)$$

$$= I_S R e^{\left(\frac{-V_{in}}{V_T}\right)}$$

## 2. Block Diagram



# Exercise (2/5)

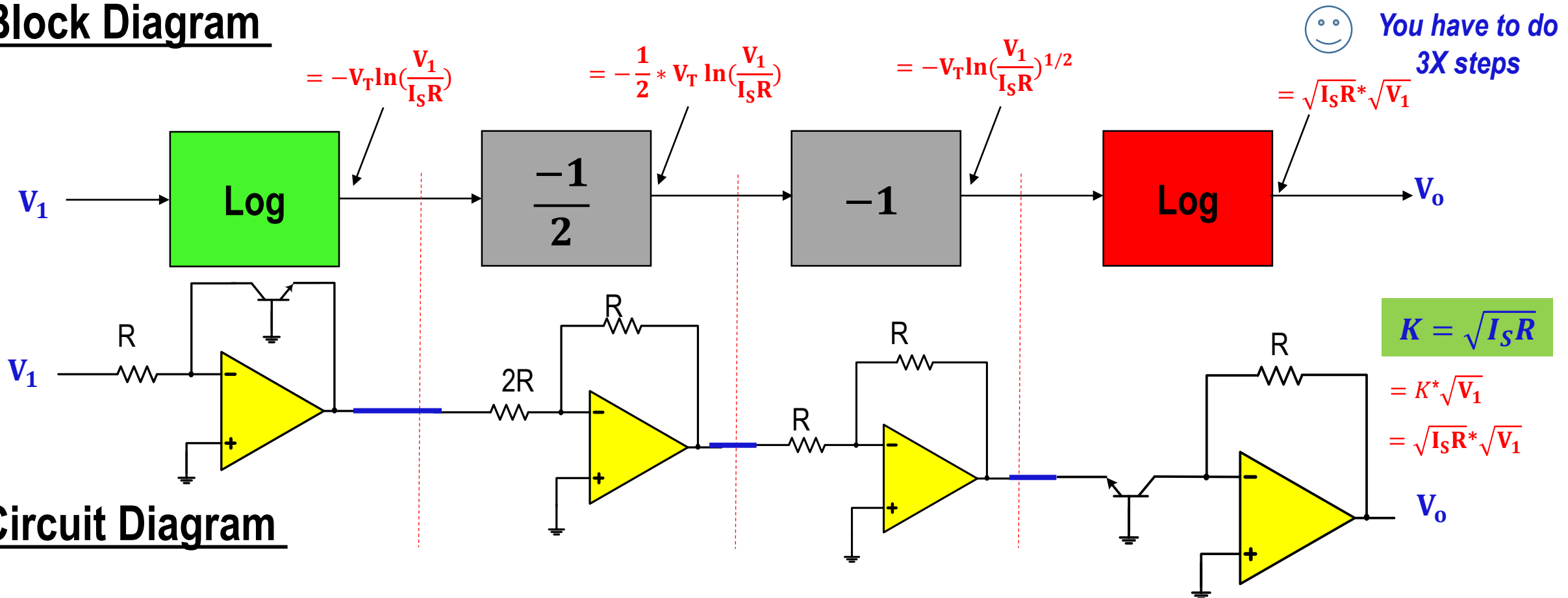


□ Design using Op. Amp a circuit to realize the following function

(a)  $V_o = K \sqrt{V_1}$ , where  $K$  is constant

Ans.

## 2. Block Diagram



## 3. Circuit Diagram

# Exercise (3/5)



□ Ex.2, Design using Op. Amp a circuit to realize the following function

(a)  $V_o = K \frac{V_1}{V_2}$ , where  $K$  is constant

Ans.

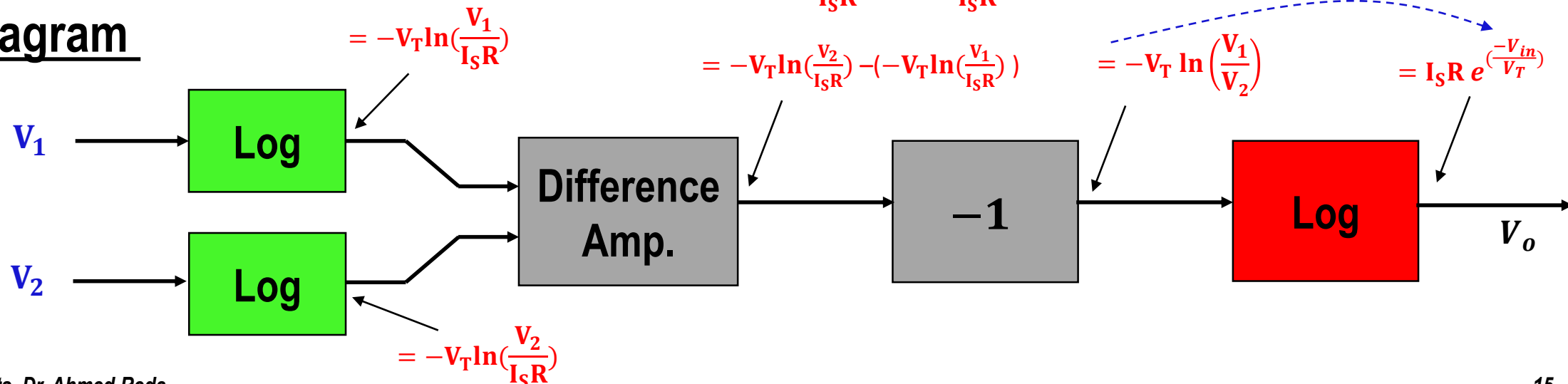
## 1. Refine the function

$$V_o = k \frac{V_1}{V_2}$$

$$V_o \propto \frac{V_1}{V_2}$$

$$\ln(V_o) \propto \ln(V_1) - \ln(V_2)$$

## 2. Block Diagram



You have to do 3X steps

$$= K * \frac{V_1}{V_2}$$

$$= I_S R * \frac{V_1}{V_2}$$

$$= I_S R e^{\ln\left(\frac{V_1}{V_2}\right)}$$

$$= I_S R e^{\left(\frac{-V_{in}}{V_T}\right)}$$

# Exercise (4/5)

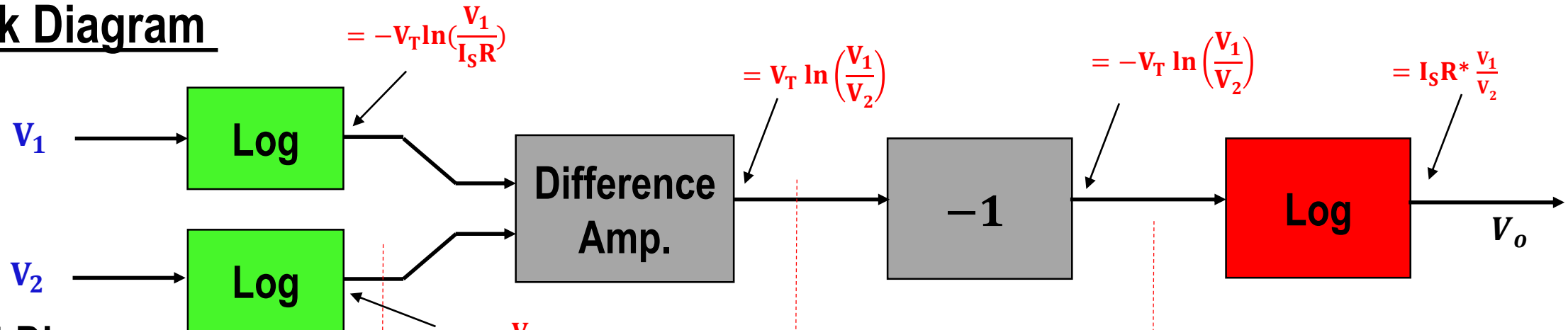


□ Design using Op. Amp a circuit to realize the following function

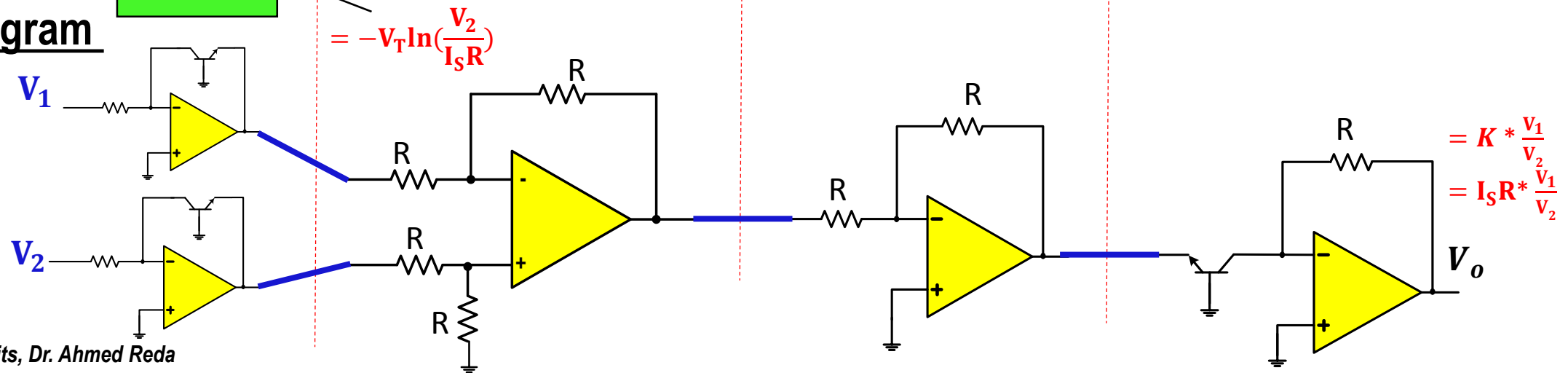
(a)  $V_o = K \frac{V_1}{V_2}$ , where  $K$  is constant

Ans.

## 2. Block Diagram



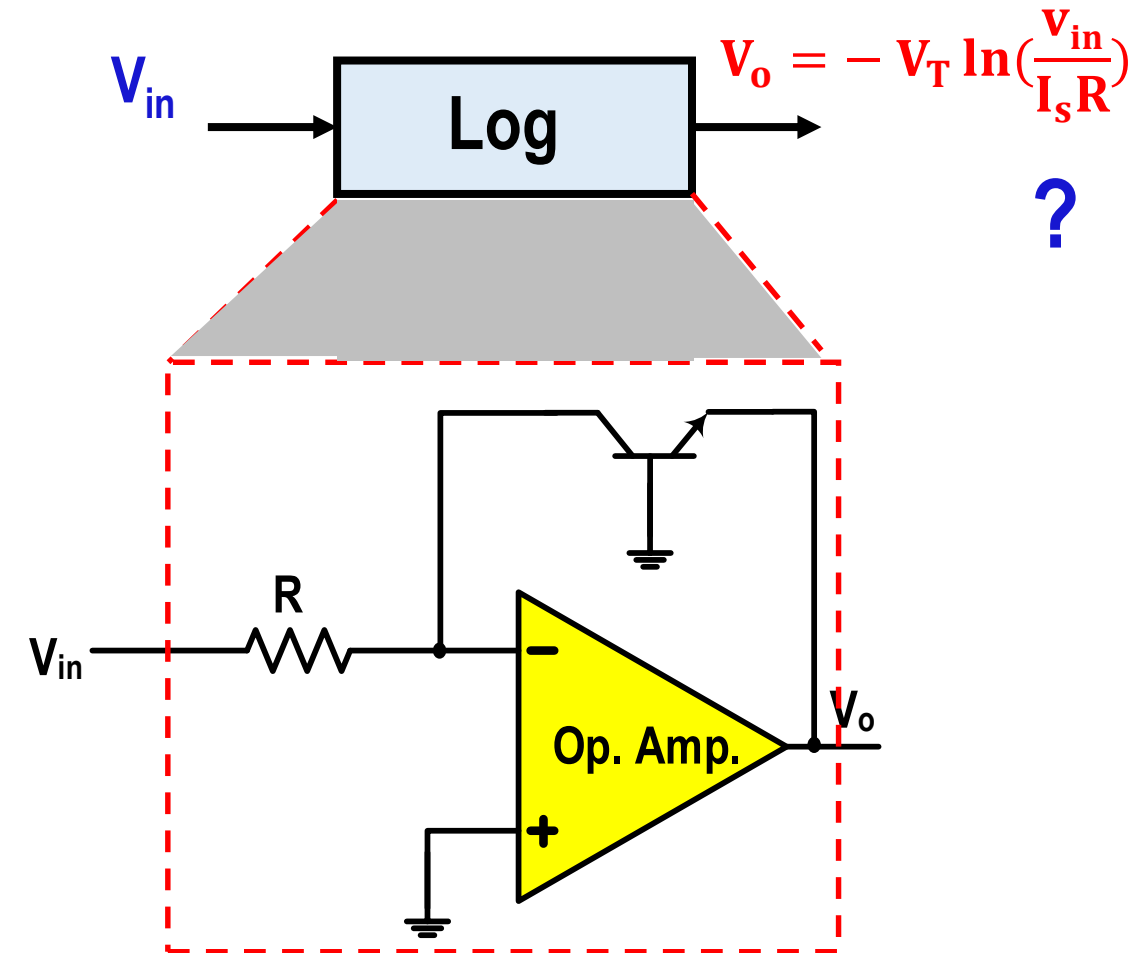
## 3. Circuit Diagram



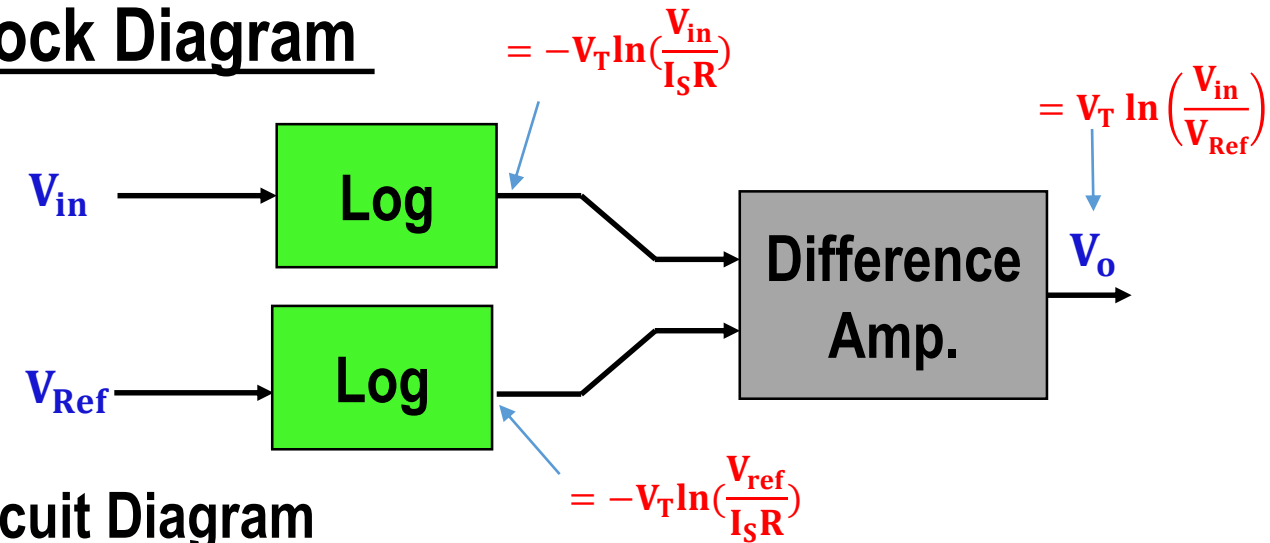
# Exercise (5/5)

□ Ex.,3 Design a log amplifier which is independent on “Is”

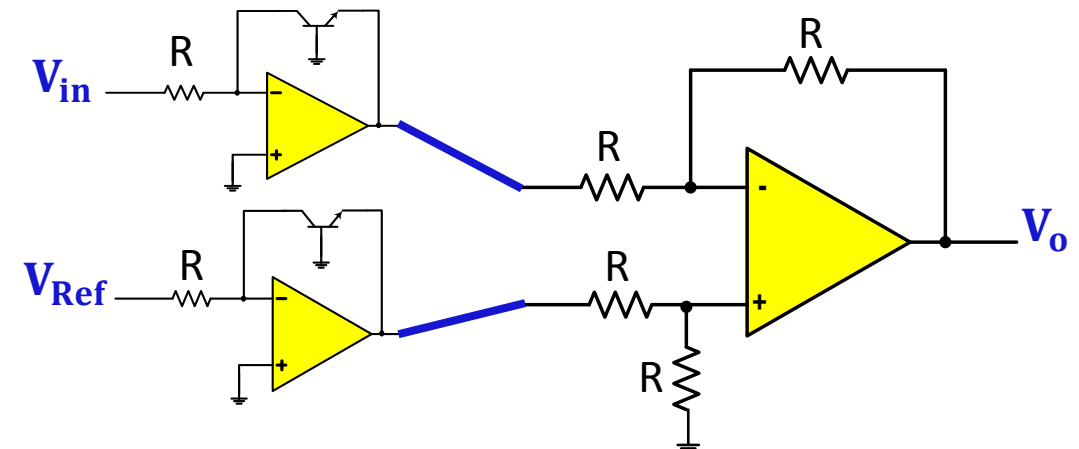
Ans.



## Block Diagram



## Circuit Diagram



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# Thanks!